Self-Improvement Agent Harness: A Deterministic SIA Exemplar

Meta → Target → Feedback loops with public/private task splits

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1 Abstract

This exemplar documents **template_sia**, a deterministic implementation of the Self-Improvement Agent (SIA) harness contract described in [AI, 2026]. The default pipeline replays fixture-backed generations for the `mini_classify` task; opt-in live mode runs bounded target subprocesses and optional Ollama-backed meta/feedback steps.

Run snapshot. Task `mini_classify`, run 1, 3 generation(s), live=false. Final accuracy=1.0000 over 6 held-out samples. Values are injected by `scripts/z_generate_manuscript_variables.py` after analysis.

Keywords: self-improvement agents, benchmark harness, reproducible evaluation, agent loops

2 Introduction

This exemplar ships **template_sia**, a deterministic research harness for Self-Improvement Agent (SIA) loops [AI, 2026]. It documents how the template repository separates generic orchestration (`infrastructure/sia/`) from a reproducible project surface (`projects/templates/template_sia/`) without vendoring the upstream upstream SIA orchestrator repository.

Compared with the AutoResearch exemplar, SIA focuses on **meta** → **target** → **feedback** generations with public/private task splits rather than candidate-model search and readiness gates. Default CI replays fixture-backed generations; live mode remains opt-in.

Scope. The bundled `mini_classify` task is a threshold classifier on one feature column. Results demonstrate harness wiring and artifact contracts—not state-of-the-art accuracy.

Run contract. Task `mini_classify`, run 1, up to 3 generation(s), live=false.

3 Methodology

3.1 SIA loop

The harness implements a three-agent cycle:

1. **Meta** — proposes or seeds a target agent for generation n.
2. **Target** — runs against public task data and writes `agent_execution.json`.
3. **Feedback** — reads private evaluation metrics and proposes improvements for generation $n + 1$.

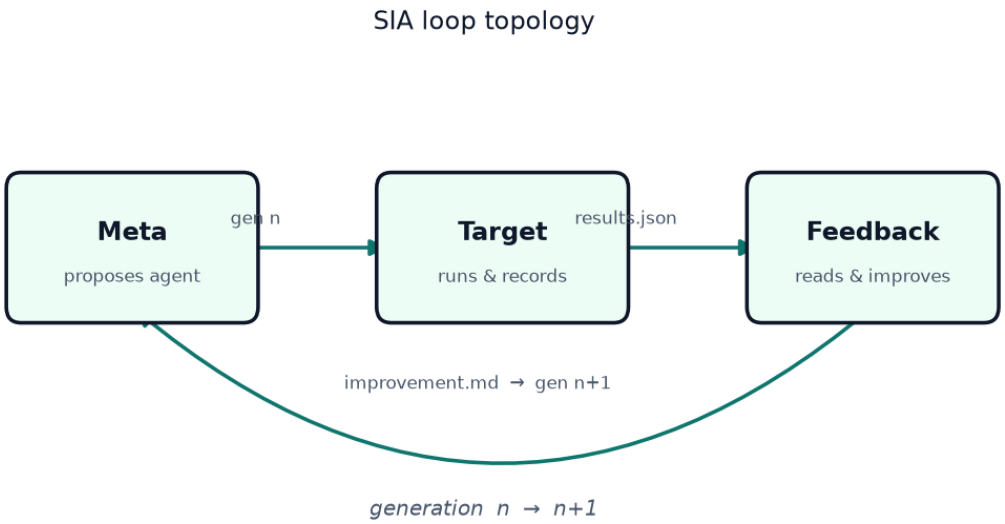


Figure 1: Meta → Target → Feedback loop topology for the SIA harness, generated programmatically by `write_sia_loop_topology`.

Artifacts land under `output/runs/run_{id}/gen_{n}/` with `target_agent.py`, `agent_execution.json`, optional `improvement.md`, and canonical `results.json`.

3.2 Task layout

Each task exposes:

- `data/public/` — agent-visible instructions and data (`task.md`, `train.csv`, `evaluate.py`).
- `data/private/` — held-out labels for evaluation only.
- `reference/reference_target_agent.py` — deterministic baseline.

The exemplar task `mini_classify` is a threshold classifier on a single feature column.

3.3 Determinism contract

When `sia.live: false` (default), generations replay recorded fixtures from `src/fixtures/recorded_generations/`. CI never executes generated agent code or calls external LLM APIs.

Pass `--live-sia` to `scripts/run_sia_loop.py` for bounded subprocess execution and optional Ollama feedback when a model is configured.

3.4 Per-generation metric overview

The heatmap below provides a compact overview of per-generation accuracy and sample count, enabling at-a-glance comparison across runs.

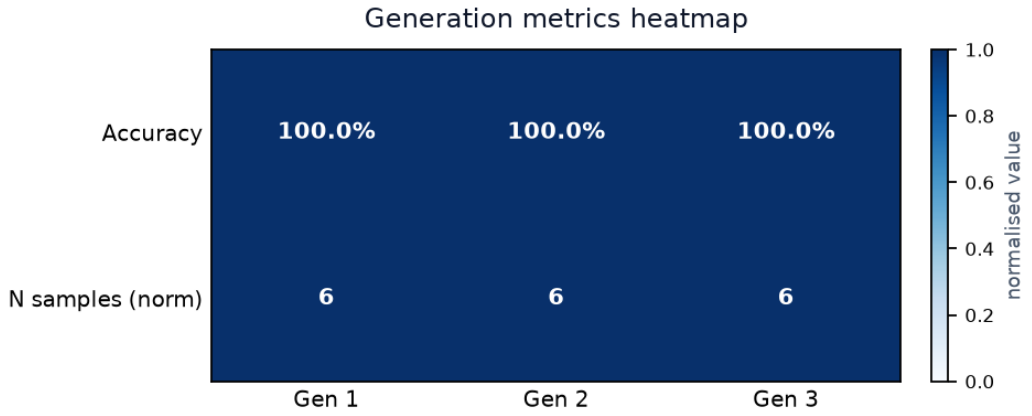


Figure 2: Per-generation metric heatmap showing accuracy and sample count across SIA generations.

4 Results

tbl. 1 summarizes fixture-replay metrics for the bundled run.

Table 1: SIA generation metrics (fixture replay).

Gen	Metric	Value	N
1	accuracy	1.0000	6
2	accuracy	1.0000	6
3	accuracy	1.0000	6

Metric delta (final − first generation): 0.0000.

Final injected token: accuracy=1.0000 (n=6).

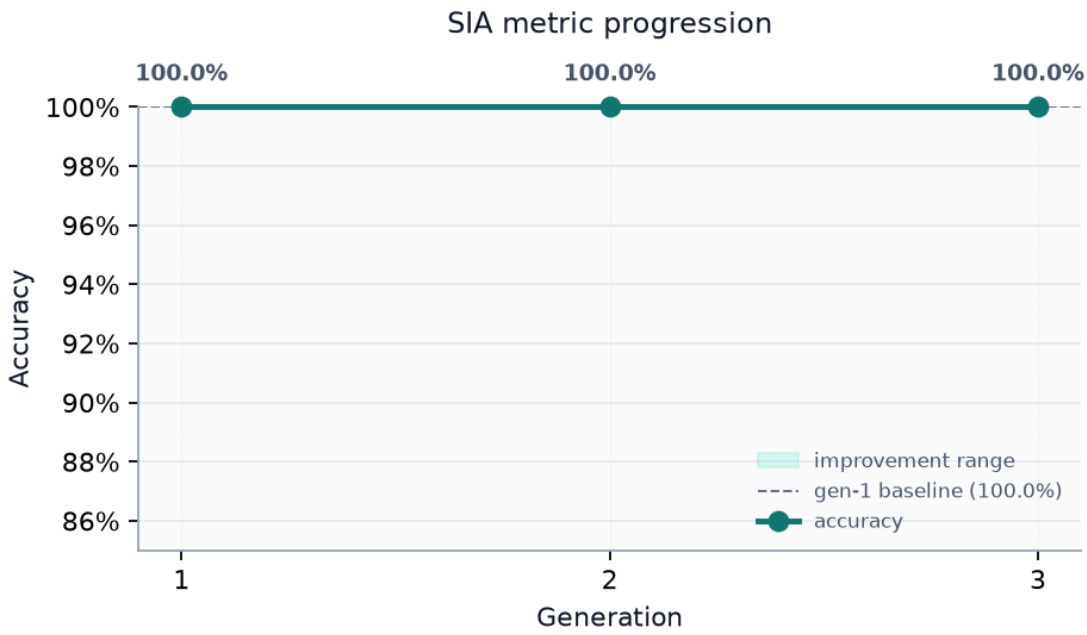


Figure 3: SIA metric progression across generations.

4.1 Threshold robustness control

The recorded target agents genuinely apply the proposed thresholds 0.5, 0.3, and 0.25. All three lie inside this toy dataset's clean separation gap and therefore score 1.0. The zero generation-over-generation delta is an important negative result: the harness records feedback and code variation without fabricating an improvement signal where the evaluation cannot distinguish one.

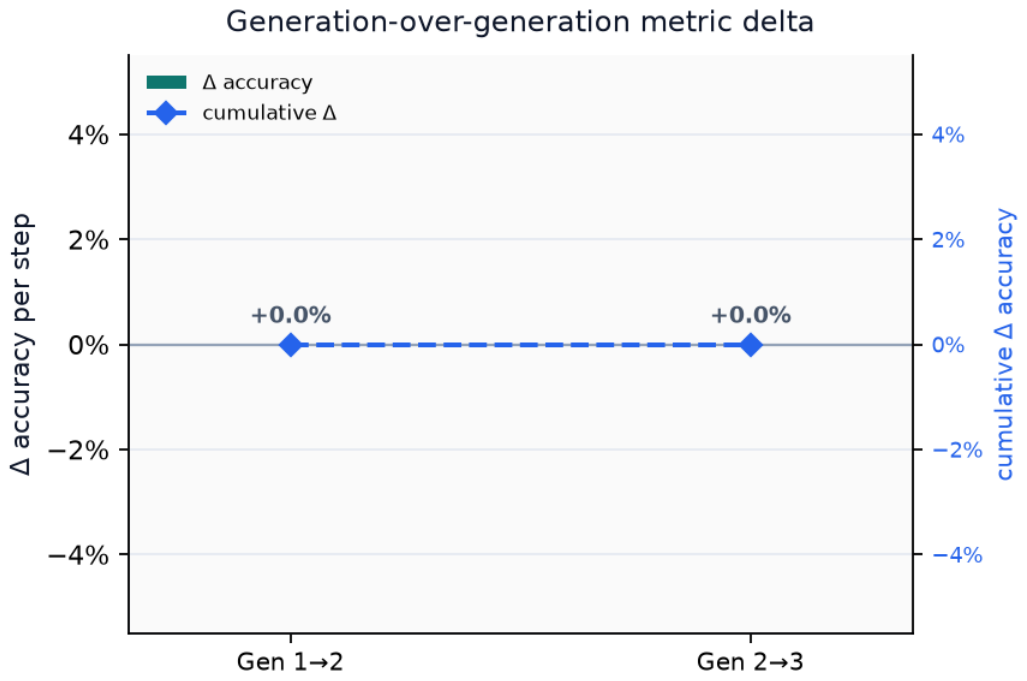


Figure 4: Generation-over-generation metric delta (Δaccuracy) for the SIA fixture replay; the flat trace confirms threshold robustness and prevents a fabricated improvement claim.

5 Conclusion

template_sia demonstrates how to embed the SIA harness contract in the Research Project Template without vendoring upstream orchestration code. Layer 1 (`infrastructure/sia/`) owns task validation, evaluation, context logging, and the generation state machine; Layer 2 wires a minimal classification task, fixture replay, and manuscript tokens.

Non-claims. This tree is a harness and documentation exemplar. Fixture-replay metrics (final accuracy=1.0000) validate wiring only—they are not evidence of state-of-the-art self-improvement. Live self-modification and external LLM calls remain opt-in; default CI never executes generated agent code or claims benchmark parity with [AI, 2026].

6 References

See `references.bib` for BibTeX entries cited in this manuscript, including [AI, 2026] and the template repository DOI from `manuscript/config.yaml`.

Hexo AI. Self-improvement agents. 2026. URL https://arxiv.org/abs/2605.27276.